COMPARATIVE ANALYSIS OF DECODER-ONLY ARCHITECTURES FOR ABSTRACTIVE TEXT SUMMARIZATION

Pretraining vs Training from Scratch

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INTRODUCTION

What is Natural Language Processing?

is a branch of artificial intelligence that enables computers to understand, interpret, and generate human language.

Shift toward deep learning approaches

Modern NLP has shifted toward deep learning models, which outperform traditional methods by capturing complex linguistic patterns and context.

Importance of text summarization

Text summarization condenses large information into concise, meaningful insights.

WHAT IS TEXT SUMMARIZATION?

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Text summarization is a Natural Language Processing (NLP) technique that condenses long documents into concise versions while preserving key information.

Extractive Summarization: Identifies and extracts important sentences or phrases from the original text to form a summary. It acts like a "highlighter".

Abstractive Summarization: Understands the content and generates new sentences, paraphrasing the original text like a human would.

PROBLEM STATEMENT

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Generating human-like summaries is difficult

The model must understand the core meaning of the entire text to rephrase it accurately.

Limited computational resources

It involves complex, deep learning, encoder-decoder architectures, such as Transformer models, which are harder to train than extractive models.

“Generating human-like summaries is challenging due to the need for deep understanding and computationally intensive models.”

OBJECTIVES

01 Compare pretrained vs scratch model

- Analyze how models initialized with prior knowledge differ from those trained entirely on task-specific data.

02 Evaluate performance of decoder-only models

- Assess the effectiveness of decoder-only architectures in generating coherent and accurate summaries.

03 Understand impact of pretraining

- Examine how pretraining influences model performance, convergence speed, and summary quality.

BACKGROUND – TRANSFORMER MODELS

Transformer is the core architecture behind modern AI, powering models like ChatGPT, and Gemini

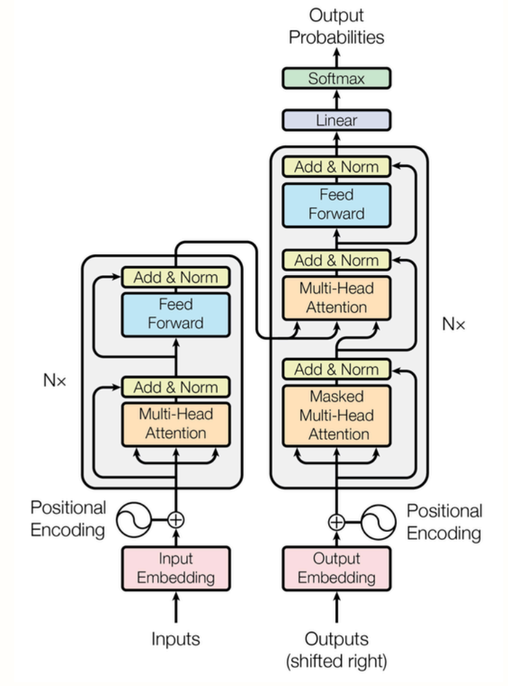
Transformer was first introduced in the seminal paper "Attention is All You Need" in 2017 and has since become the go-to architecture for deep learning models, powering text-generative models like OpenAI's GPT, Meta's Llama, and Google's Gemini

BACKGROUND – HOW TRANSFORMERS WORK

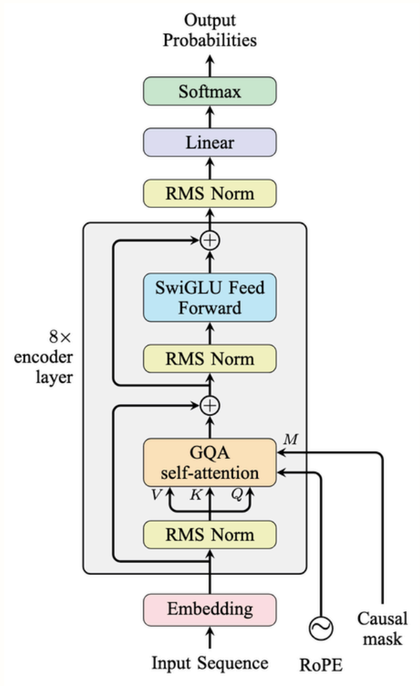
Trasformers aren’t magic - They build text step by step by asking:

“What is the most probable next word that will follow this input”

ENCODER VS DECODER ARCHITECTURES



Encoder-Decoder Architecture



Decoder-Only Architecture

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GPT-2 OVERVIEW

GPT-2 is a large-scale language model pre-trained on diverse internet text to generate coherent and contextually relevant language.

01 **Autoregressive language model:** It is an autoregressive model that generates text by predicting each token based on previously generated tokens.

02 **Predicts next word sequentially:** The model produces text step-by-step by sequentially predicting the next word in a sequence.

03 **Decoder-only transformer architecture:** It uses a decoder-only Transformer architecture, relying solely on masked self-attention to model context without an encoder.

DATASET USED

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CNN/DailyMail dataset:

CNN/DailyMail is a widely used dataset for training and evaluating text summarization models.

News articles + summaries:

It consists of news articles paired with human-written summaries for supervised learning.

Standard benchmark for summarization:

It serves as a standard benchmark to compare the performance of different summarization approaches.

SYSTEM OVERVIEW

Two models:

- Trained from scratch,
- Pretrained, fine-tuned

- The system uses two models, one trained from scratch and another pretrained then fine-tuned, for comparison.

- Goal: The goal is to evaluate and compare their performance in generating accurate and coherent summaries.

ATTENTION MECHANISM

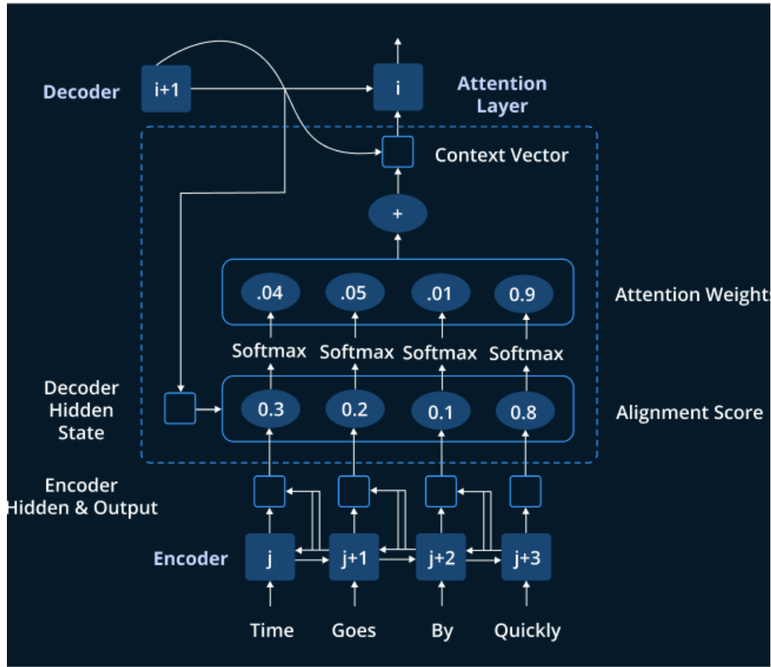
Focus on relevant words:
The attention mechanism allows the model to focus on the most relevant words in a sequence when generating output.

Query, Key, Value (QKV)

It computes relationships between words using Query, Key, and Value vectors to determine importance.

Enables context understanding

This mechanism helps the model capture long-range dependencies and understand context effectively.

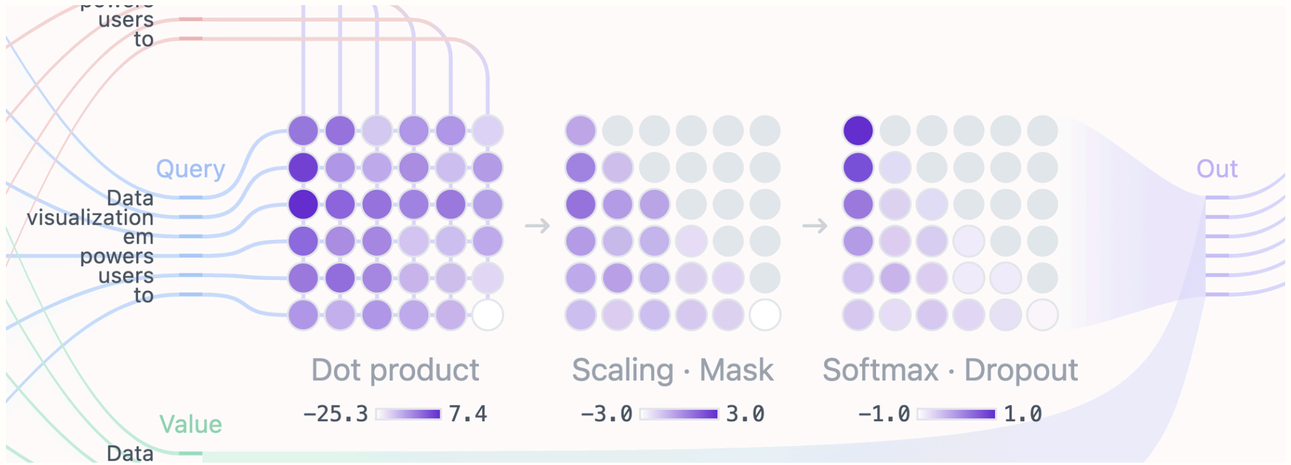


ATTENTION CONT...

- Query: what we're looking for
  - The Query represents the information the model is currently seeking.

- Key: what we compare against
  - The Key represents the features used to match and measure relevance with the Query.

- Value: actual information
  - The Value contains the actual content that is passed forward based on attention scores.



CAUSAL SELF-ATTENTION

- Causal self-attention is a mechanism in decoder-only models where each token can only attend to previous tokens (not future ones), ensuring the model generates text in a left-to-right sequence.

- **Only looks at previous tokens**: At each step, the model computes attention only over earlier words in the sequence, using a masking technique to block access to future positions.

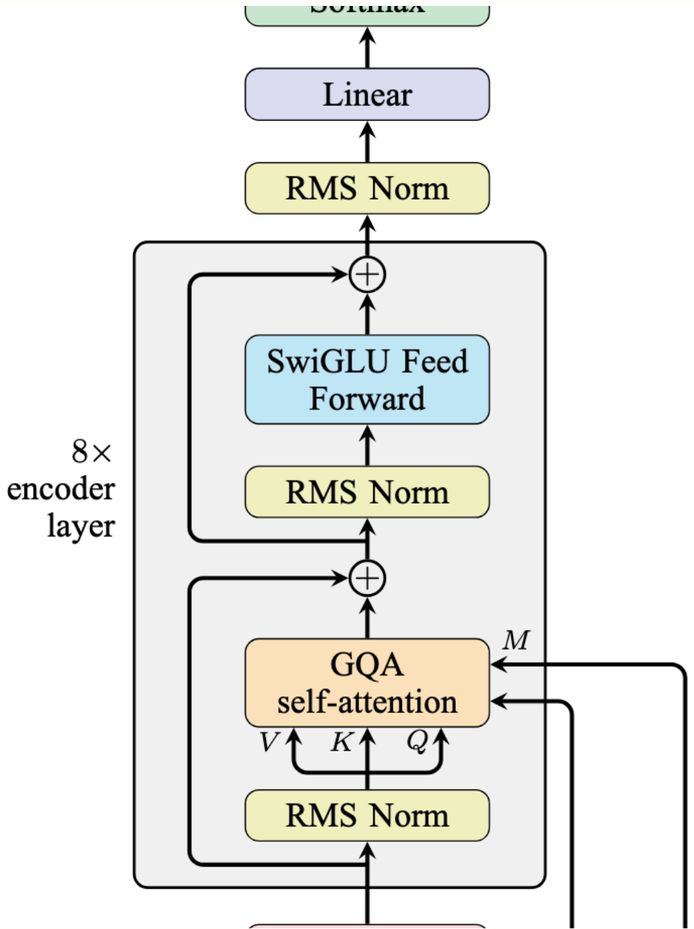
- **Prevents "future leakage":** This masking prevents information from upcoming tokens from influencing predictions, which would otherwise make training unrealistic and invalidate generation behavior.

VISUALIZER

MODEL ARCHITECTURE DETAILS

1. **Token embeddings**
2. **Positional encoding**
3. **Transformer blocks**
4. **Layer normalization**

Together, these components allow the model to understand context, sequence, and meaning for generating high-quality summaries.



1. TOKEN EMBEDDINGS

- Converts words/subwords into dense numerical vectors.
- Each token → fixed-size vector (e.g., 768 dimensions in GPT-2)
- Captures semantic meaning (similar words → similar vectors)
- Learned during training

2. POSITIONAL ENCODING

- Transformers don’t understand word order by default
- Adds position information to each token embedding
- Ensures the model distinguishes:
  - “dog bites man” vs “man bites dog”
- In decoder-only models like GPT-2, position is learned (not fixed sinusoidal)

3. TRANSFORMER BLOCKS

- Core building unit (stacked multiple times, e.g., 12 layers in GPT-2)
- Each block contains:
  - Masked Multi-Head Self-Attention
    - Looks at previous tokens only (causal)
    - Captures relationships between words
  - Feedforward Neural Network (FFN)
    - Applies non-linear transformations
- Multi-head attention allows the model to:
  - Focus on different relationships simultaneously

4. LAYER NORMALIZATION

- Stabilizes and speeds up training
- Normalizes outputs within each layer
- Helps prevent exploding/vanishing gradients
- Applied multiple times inside each transformer block

Thank You!